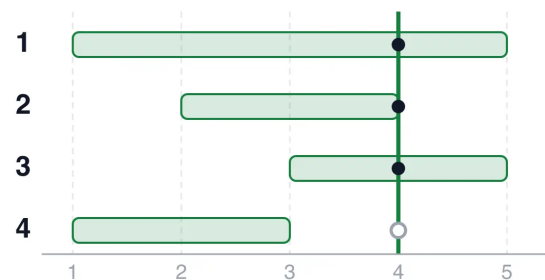
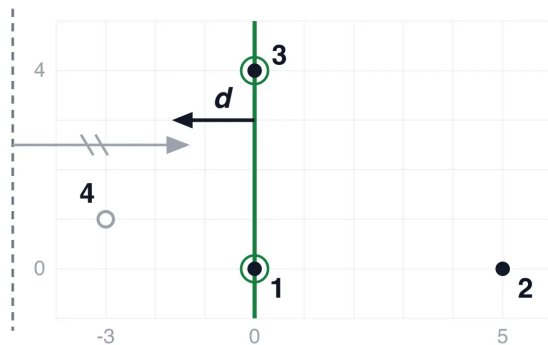


Jammers

A spectrum monitoring center watches a flat region in which N sources of radio interference, numbered 1 through N , have been located. Source i sits at the point (x_i, y_i) and transmits continuously from minute L_i through minute R_i , inclusive. Source i is **active** at minute t when $L_i \leq t \leq R_i$.

The monitoring works like this. The center owns a steerable antenna array. One measurement is described by a minute t and a non-zero direction (d_x, d_y) . The measurement reports the active source for which $x_i \cdot d_x + y_i \cdot d_y$ is as large as possible. If several active sources attain that maximum, the measurement reports the one with the smallest index.

Equivalently, imagine a straight line perpendicular to (d_x, d_y) , placed so far away in the direction (d_x, d_y) that every source lies behind it. The line now sweeps backwards, staying perpendicular to (d_x, d_y) , until it first touches a source that is active at minute t . The measurement reports that source.



Both pictures show the example below. On the left, the four sources at minute 4 measured in the direction $(-1, 0)$: the dashed line is the sweeping line before it moves, and it travels the opposite way to d . Source 4 is drawn hollow because it is not active at minute 4, so the line passes over it; the line stops at sources 1 and 3 together, and the smaller index is reported. On the right, the minutes during which each source transmits, with the five measured minutes along the bottom and minute 4 marked.

Answer Q measurements.

Input format

The first line contains an integer N .

Each of the next N lines contains four integers x_i, y_i, L_i , and R_i , describing source i .

The next line contains an integer Q .

Each of the next Q lines contains three integers t, d_x , and d_y , describing one measurement.

Output format

Print Q lines. The k -th line must contain the index of the source reported by the k -th measurement, in input order.

Constraints

- $1 \leq N, Q \leq 200\,000$
- $-10^9 \leq x_i, y_i \leq 10^9$
- $1 \leq L_i \leq R_i \leq 10^9$
- $1 \leq t \leq 10^9$, and at least one source is active at every minute that is measured
- $-10^9 \leq d_x, d_y \leq 10^9$ and $(d_x, d_y) \neq (0, 0)$
- The N positions are pairwise distinct, and no three of them lie on a common line.

Subtasks

Subtask	Points	Additional constraints
1	5	$N, Q \leq 500$; $L_i = 1$ and $R_i = 10^9$ for every source; $1 \leq x_i, y_i, d_x, d_y \leq 10^4$; and every measurement has a unique maximizing source.
2	7	$N, Q \leq 2000$
3	13	$L_i = 1$ and $R_i = 10^9$ for every source
4	15	$(d_x, d_y) = (1, 0)$ for every measurement
5	20	$N, Q \leq 60\,000$
6	40	No additional restrictions.

Examples

Example 1

Input:

```
4
0 0 1 5
5 0 2 4
0 4 3 5
-3 1 1 3
5
1 1 0
2 1 0
3 0 1
4 -1 0
5 0 1
```

Output:

```
1
2
3
1
3
```

The direction $(1,0)$ asks for the largest x_i , the direction $(-1,0)$ for the smallest, and $(0,1)$ for the largest y_i .

At minute 1 only sources 1 and 4 are active, at $(0,0)$ and $(-3,1)$; the larger x is 0, so source 1 is reported. At minute 2 source 2 has joined, and its $x = 5$ is the largest. At minute 3 all four sources are active, and source 3 has the largest y .

The fourth measurement shows the tie-break. At minute 4 sources 1, 2, and 3 are active, and the direction $(-1,0)$ gives them the values 0, -5 , and 0. Sources 1 and 3 both attain the maximum, so the smaller index is reported. This is the measurement drawn in the figure above.